

PUNEETH JOSEPH

Principal Software Engineer

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PROFESSIONAL SUMMARY

Principal Software Engineer with 14+ years of experience specializing in automotive operating systems and ADAS Domain Controller development. Led development and integration of MB.OS (Mercedes-Benz Operating System), a Linux and QNX-based platform enabling software-driven architecture. Expert in software-driven vehicle architecture integration, Adaptive AUTOSAR, QNX, Linux, and embedded systems architecture. Proven track record at Mercedes-Benz Group and Robert Bosch, delivering next-generation software-driven automotive platforms integrating multiple sensor technologies (Cameras, Radar, Lidar, Ultrasonic). Strong background in software architecture, system integration, and cross-functional collaboration with platform teams, application developers and external partners.

TECHNICAL SKILLS

Automotive Software Architecture

- MB.OS (Mercedes-Benz Operating System)
- Software-Driven Architecture
- Adaptive AUTOSAR, AUTOSAR Classic
- QNX, Linux, POSIX Systems
- Service-Oriented Architecture (SOA)
- Embedded Software Architecture

Programming & Scripting

- C++, C
- Shell Scripting
- Python (automation & tooling)

Development Tools & Platforms

- QNX Momentics IDE
- Nvidia Drive Platform (Orin SoC)
- Debugging Tools (GDB, JTAG)
- CI/CD Tools
- Git, SVN
- Hypervisor Technologies

Domain Expertise

- MB.OS Development & Integration
- ADAS Domain Controllers
- ECU Development & Integration
- Virtual ECU (vECU) Development
- Image Processing (CNN, SSD)
- Sensor Integration
- Navigation Systems Middleware
- Over-the-Air (OTA) Updates
- Level 3 Autonomous Driving Systems

PROFESSIONAL EXPERIENCE

Principal Software Integrator

July 2021 - October 2025

Mercedes-Benz Group

Stuttgart, Germany

Lead Software Integrator for Mercedes-Benz Operating System (MB.OS) development and ADAS Domain Controller projects, delivering next-generation software-driven architecture and platform solutions integrating Linux, QNX, AUTOSAR Classic, and AUTOSAR Adaptive systems.

- Developed and integrated MB.OS (Mercedes-Benz Operating System), a Linux and QNX-based platform for the Mercedes Modular Architecture (MMA), enabling software-driven architecture with over-the-air updates and full software stack control
- Developed and integrated multi-sensor ADAS platform supporting Cameras, Ultrasonic sensors, Radar, and Lidar technologies. Integrated dual-OS architecture (QNX for production safety-critical systems and Linux for development) and Nvidia Drive platform with Orin system-on-chip (SoC) for advanced computing capabilities, supporting Level 3 autonomous driving features
- Collaborated with platform teams, application developers, and external partners (including Nvidia) to define software architecture and integration strategies
- Maintained and released Bring-Up Software for Release Boards, enabling pre-delivery evaluation and validation of release software
- Served as Technical Contact Partner for internal and external customers, providing expertise on software integration and troubleshooting
- Customized QNX OS for custom ECU hardware from Nvidia and Qualcomm using QNX Momentics IDE
- Designed embedded software architecture for microprocessor-based automotive systems
- Implemented Service-Oriented Architecture solutions for embedded domain

Specialist Software Engineer

August 2013 - June 2021

Robert Bosch Engineering and Business Solutions

Bengaluru, India

Developed ADAS software components and middleware solutions for automotive ECUs, working on cutting-edge projects including Virtual ECU development and AI-based object detection systems.

- Developed virtual ECU (vECU) software solutions for multiple OEMs, enabling simulation and testing of real ECU functionality
- Designed and optimized Adaptive AUTOSAR-based applications for real-time object detection, processing frame data from camera sensors
- Deployed deep learning algorithms (CNN and SSD) for object detection and classification on High Performance Computing platforms
- Contributed to middleware development for navigation systems, implementing both basic and advanced navigation features
- Architected Proof of Concept (POC) solution integrating Android, Telematics (QNX), and Bosch Navigation System (Linux) using Hypervisor technology

Software Developer

August 2010 - July 2013

Wipro Technologies

Bengaluru, India

Led platform migration project, porting EVDO application software from Solaris to Linux-based hardware platform.

- Successfully ported C++ application codebase and shell scripts from Solaris to Linux platform
- Implemented Linux-specific wrappers for shell commands, ensuring seamless functionality across platforms
- Resolved endianness compatibility issues in C++ codebase for cross-platform support
- Enhanced application features during migration process

EDUCATION

Bachelor of Engineering in Information Science

Visvesvaraya Technological University

Bengaluru, India | July 2007 - June 2010

CERTIFICATIONS

Certified Scrum Product Owner® (CSPO®)

Scrum Alliance

December 2023

ADDITIONAL INFORMATION

Languages: English (Fluent), German (Working Proficiency), Hindi (Native)

Interests: Reading, Performance Monitoring, Football, Technology Research